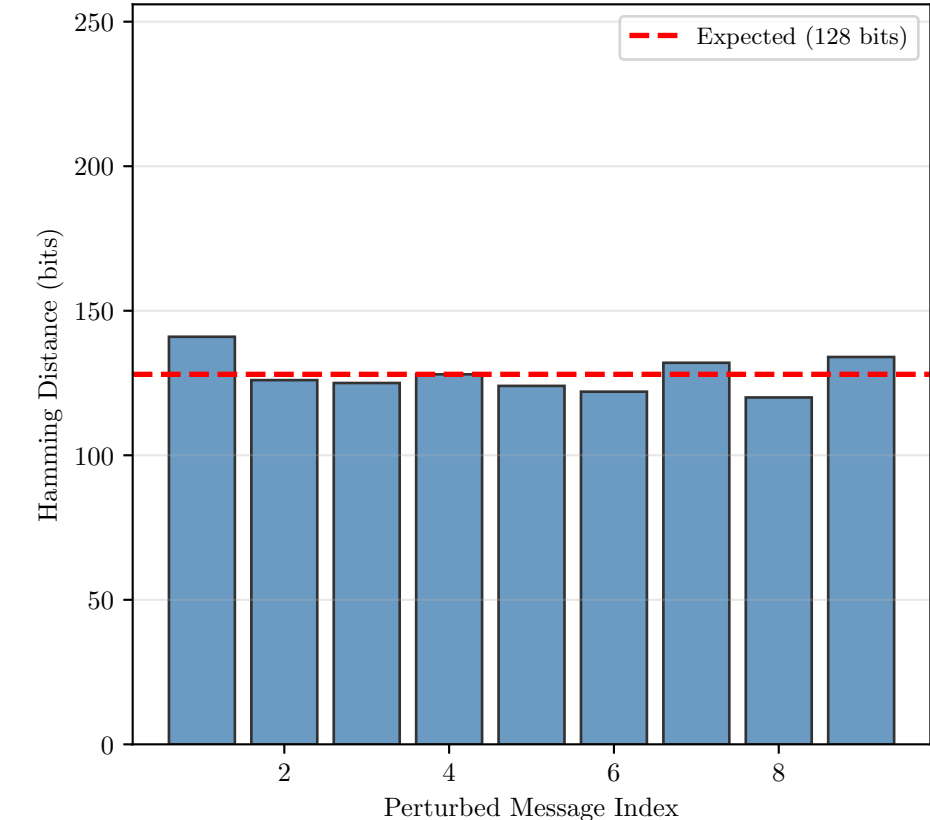
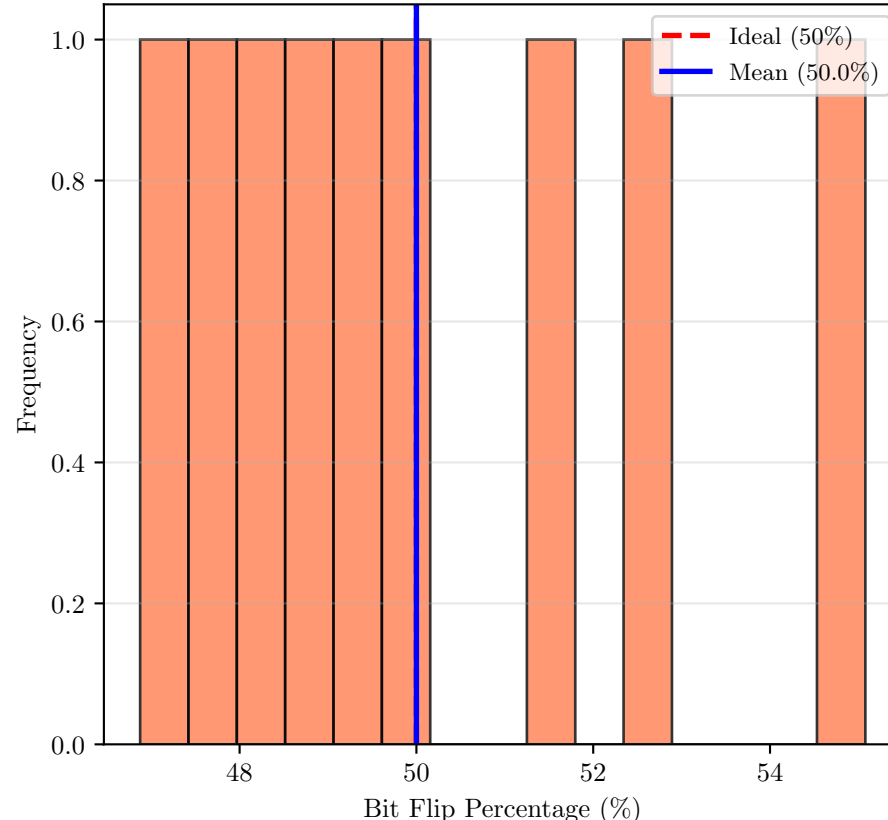


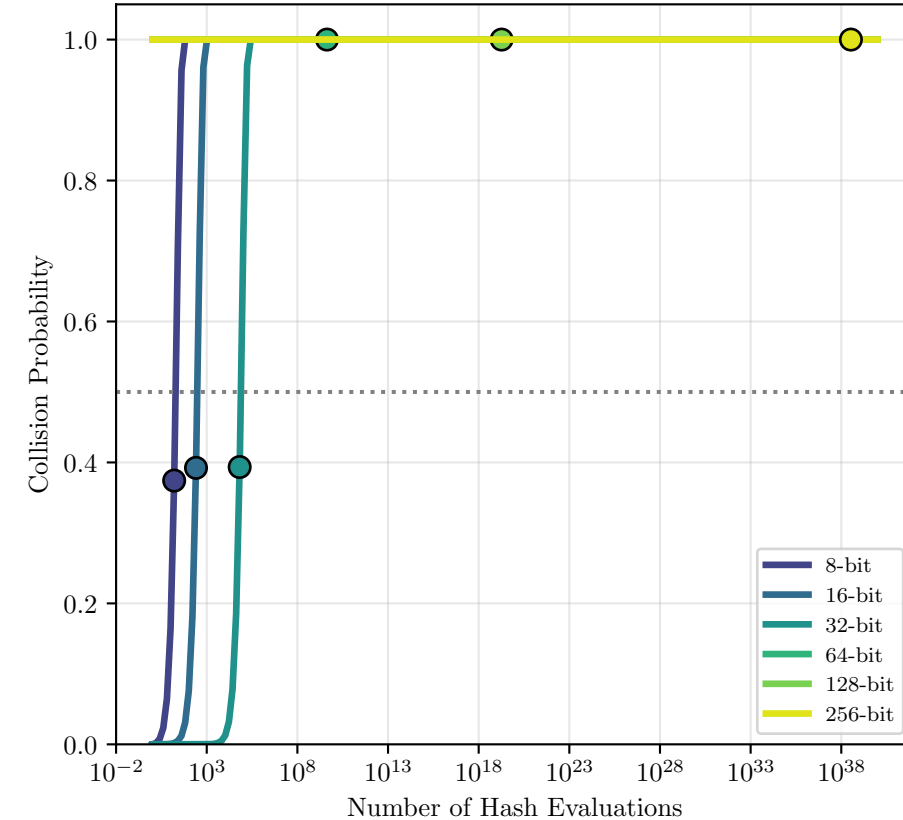
SHA-256 Avalanche Effect: Single-Bit Input Change



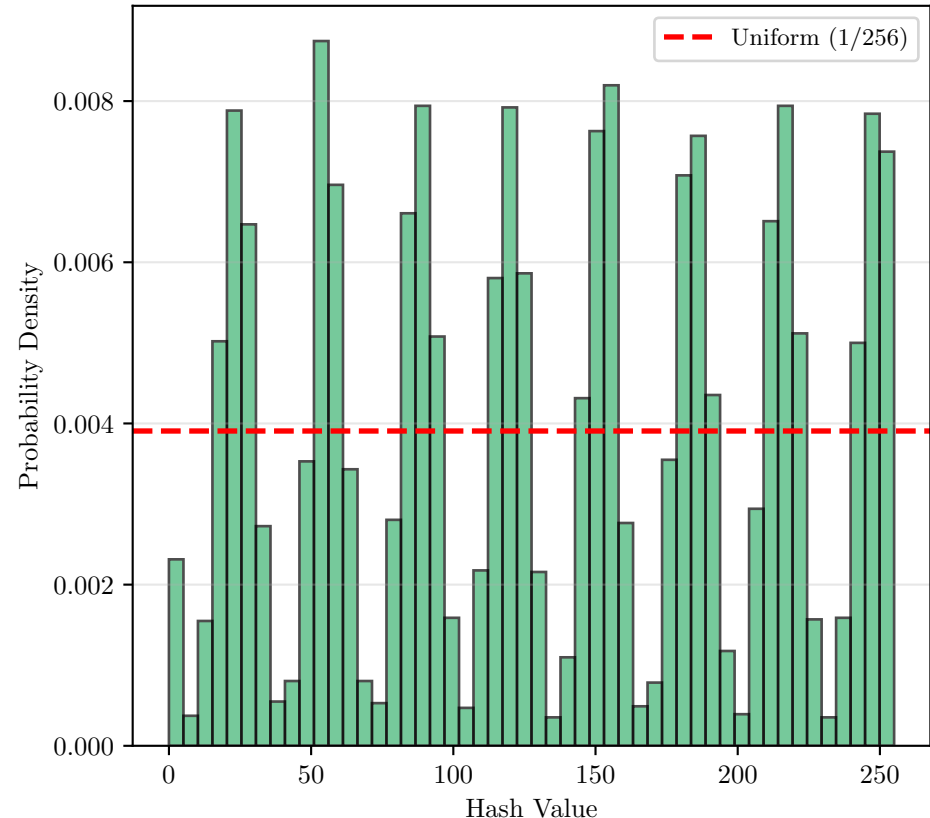
Distribution of Output Bit Changes



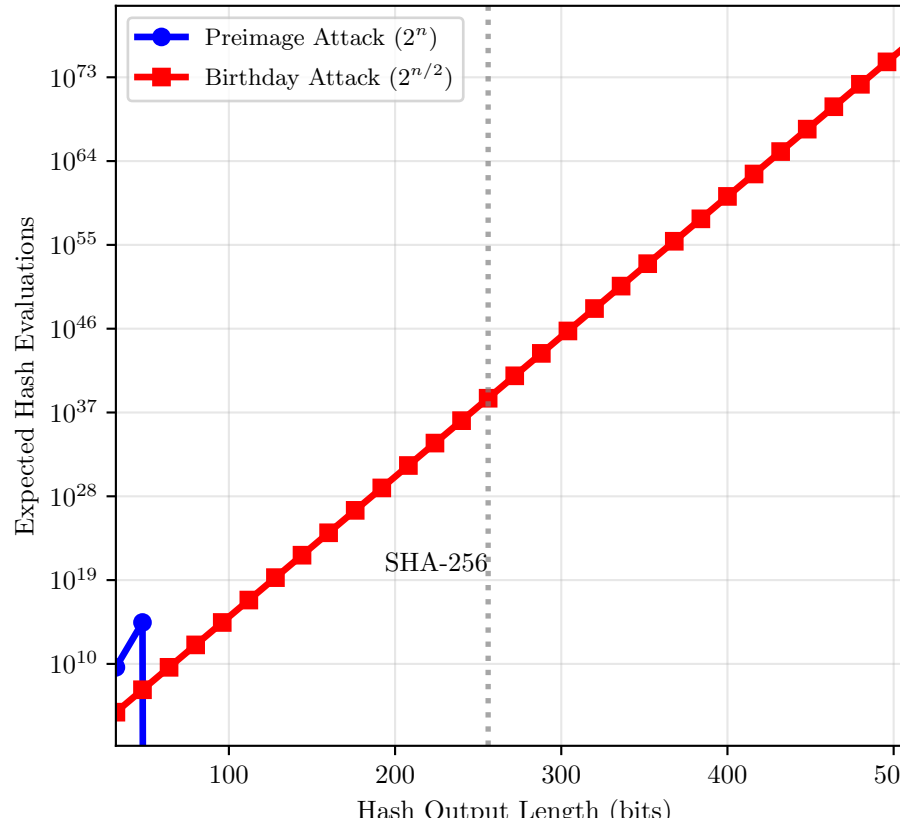
Birthday Attack: Collision Probability vs Hash Size



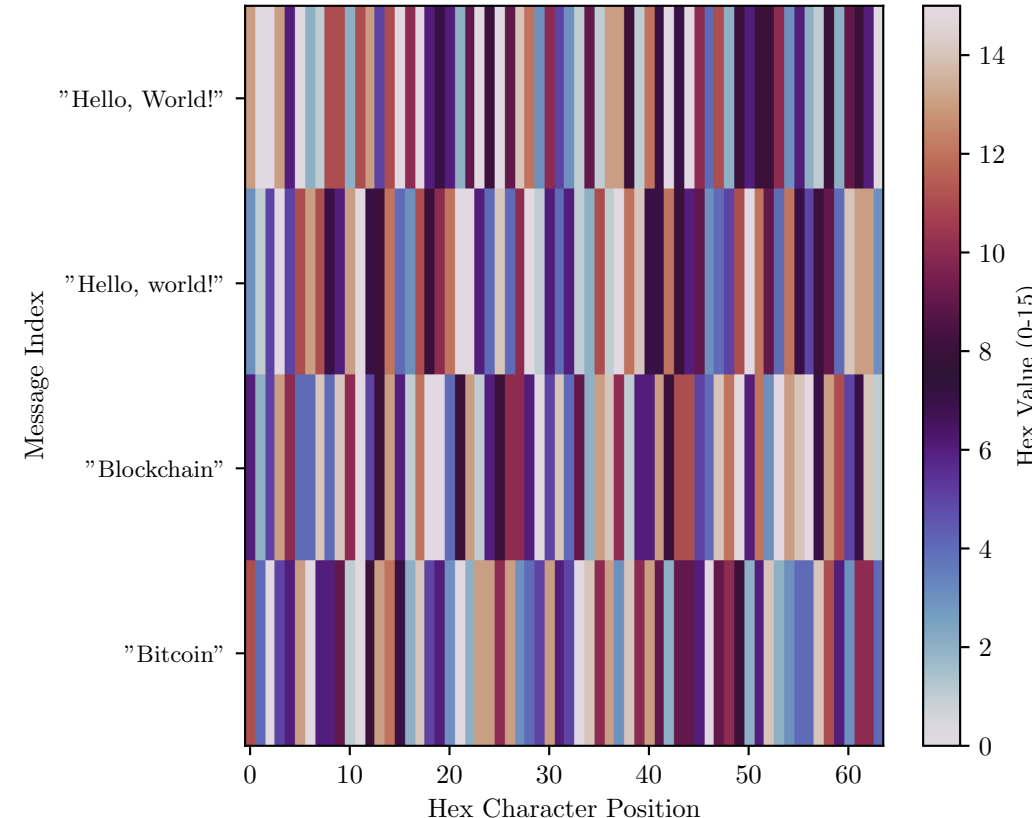
Simplified Hash Output Distribution (10k samples)



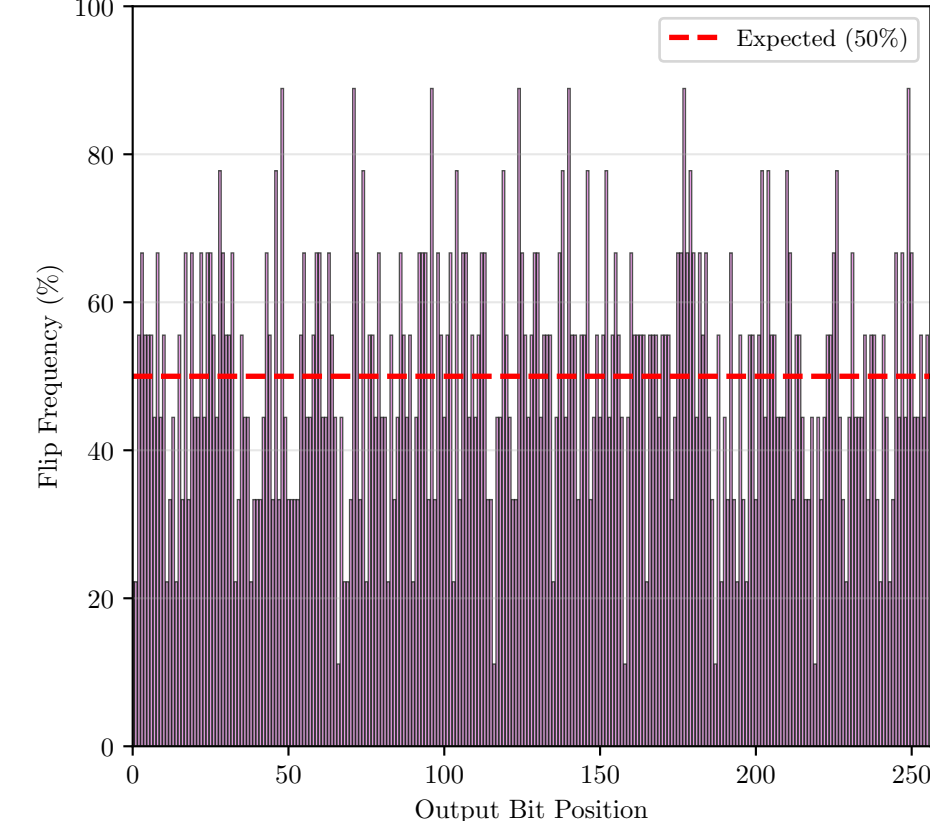
Attack Complexity vs Hash Length



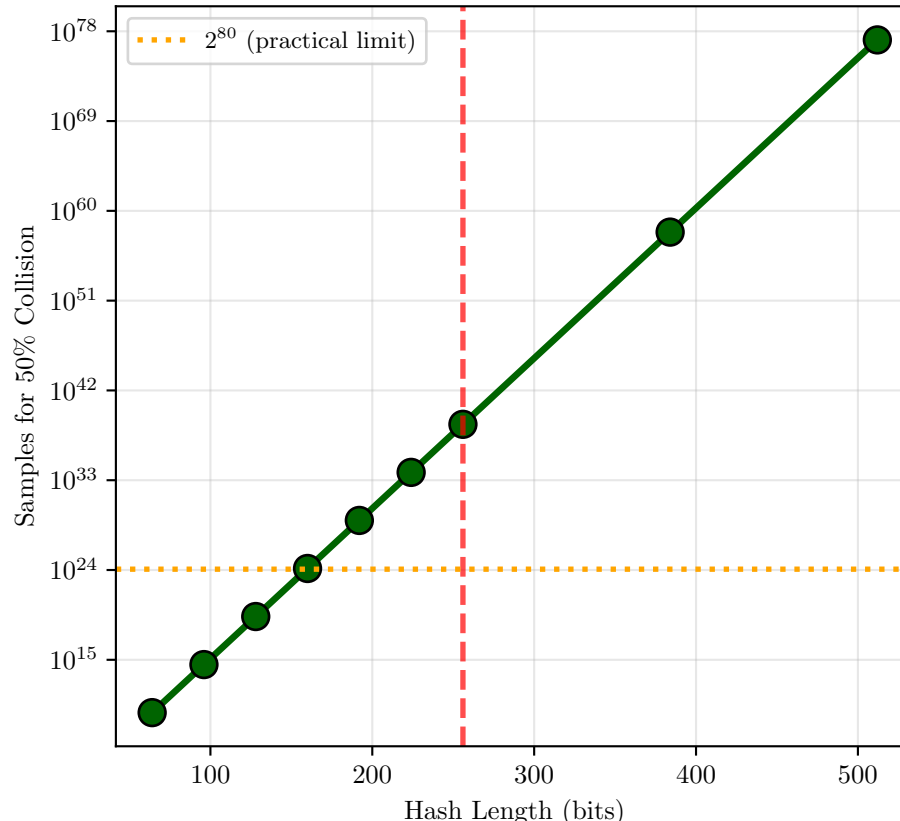
SHA-256 Digest Entropy Visualization



Per-Bit Avalanche Uniformity (9 perturbations)



Birthday Bound: Required Hash Evaluations



Fixed-Length Output Property

